FINAL PROJECT

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1. Abstract

The objective of this project is to analyze 20 years of trends in the movie theater industry through monthly movie audience data. Data is explained through time series data analysis using the ARIMA model, and it is diagnosed whether there are any errors in the fit of the model. After that, the data is interpreted based on the derived ARIMA model and explained in connection with the trends in the movie theater industry, which is also the purpose of this project. Finally, the future prospects of the industry are evaluated by forecasting the audience numbers for the next two years.

2. Introduction

We would like to find out about the trends in the movie theater industry through monthly movie theater attendance data from 2004 to 2024. The movie theater industry was among the hardest-hit sectors during the COVID-19 pandemic. In January 2020, the first COVID-19 case was confirmed in Korea. The Korean government imposed social distance to prevent the spread of infectious diseases. This was not a temporary restriction, but continued for more than a year, leading to many movies implementing actions such as postponing unlimited releases.

OTT services, such as Netflex, were able to serve as an excellent substitute for movie theaters at this time when free going out was restricted. As movie theater attendance is restricted, OTT-exclusive content has emerged as a defining trend of the era. After the pandemic was officially declared over, the OTT industry saw sustained growth, while cinema attendance has struggled to return to pre-pandemic levels. Therefore, this project aims to examine the trends of the movie theater industry, represented by the number of moviegoers, over the past 20 years, and in particular, what changes have occurred since COVID-19 in 2020. Rather than simply judging the trends of the movie theater industry according through articles, I would like to present analysis results based on data. Furthermore, we predict what the trend of the number of audiences will be in the future.

The data used for the analysis is the data provided by the Korea Box-Office Information System (KOBIS)[1]. This data is a compilation of ticketing data for movie theaters nationwide since 2004, and information is continuously updated in real-time. Among them, the data used for the project is based on the monthly number of audiences of domestic and overseas films from January 2004 to September 2024.

1) https://www.kobis.or.kr/kobis/business/stat/them/findMonthlyTotalList.do

3. Statistical Methods

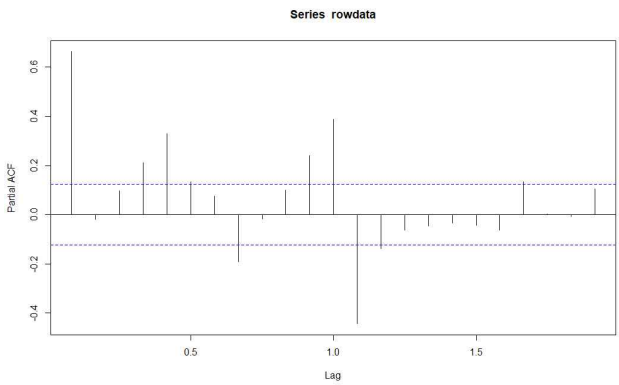
(1) Plotting the data

Data could be examined before a particular ARIMA model can be applied.

```
> rowdata
           Jan       Feb       Mar       Apr       May       Jun       Jul       Aug       Sep       Oct       Nov       Dec
2004   3071393   3113385   1844928   3307663   3817166   6668461   9451529   9835611   7597758   6254004   6032062   8243748
2005  11314789  11289736   7429408   6450931   8735786   8443990  12271728  13885001  11446609   9346940   8244955  14440828
2006  18560156  12114474   8713893   8616868  13167857  10045316  14570312  15389966   8624262  11809428   7655334  14919638
2007  15006635  12941862   9430447   7664755  12138558  11202058  17409970  20741043  11237219   8655414   9481129  16071593
2008  12851629  12844826   9606955   7330006  12870714  14227212  17033508  17370014  10944510   8744625   9472444  14131178
2009  16525596  11156303   7833062   7848673  16146532  12963286  16349119  21418298   8336580   9357091  10665552  16797532
2010  16553057  13214329   8414529   7463342  13309888  10772828  16717823  19377941  11723009   7875666   8418508  13918924
2011  12381046  13554321   8403967   7518941  13864687  12698905  18333347  20062030  13417036  12410546  10321195  16761788
2012  16628706  13065438  11392750  11993363  15970116  14211515  20964476  24238713  14597393  15288721  15947251  20592201
2013  20366773  21824737  13283201  11259977  14211968  17555754  18824441  29142663  19234287  13490048  13380122  20776436
2014  23596599  18558381  12404922   9206448  16862099  15889312  19876822  32219304  17084849  12502773  15190407  21677006
2015  22484065  16666173  11323107  12721639  17670740  14201957  23431726  30902141  18733045  14926529  15275978  18962399
2016  16908582  21312706  11266732   9995372  18131197  17020069  26237872  29940869  16208794  17159928  12682819  20161190
2017  23255258  15181395  13855445  11214167  18680206  15103839  21357163  29885346  12719893  21585909  13155930  23881676
2018  22943456  15551760  12801511  14066243  15891715  15099844  19776671  30258275  16823946  14075357  17149948  21946545
2019  18122443  22277733  14671693  13338454  18062462  22845579  21916465  24786121  14733642  14858382  18600679  22464575
2020  16843695   7372293   1834722    972572   1526236   3864510   5618701   8834699   2988647   4633609   3595309   1438873
2021   1786117   3111909   3256535   2562285   4379476   4927966   6976451   7910082   5412319   5192138   6512815   8502983
2022   5718042   3272423   2799690   3120144  14560659  15475512  16291243  14956834   9860566   6201235   6376976  14171771
2023  11250344   6421297   7476214   6972978  11747296  14522808  14280580  14564109   6656962   6900210   7642807  16700660
2024   7752095  11458714  11697143   9334928  11356606  11329904  12033799  11779803  10105373
```

<Figure 1>

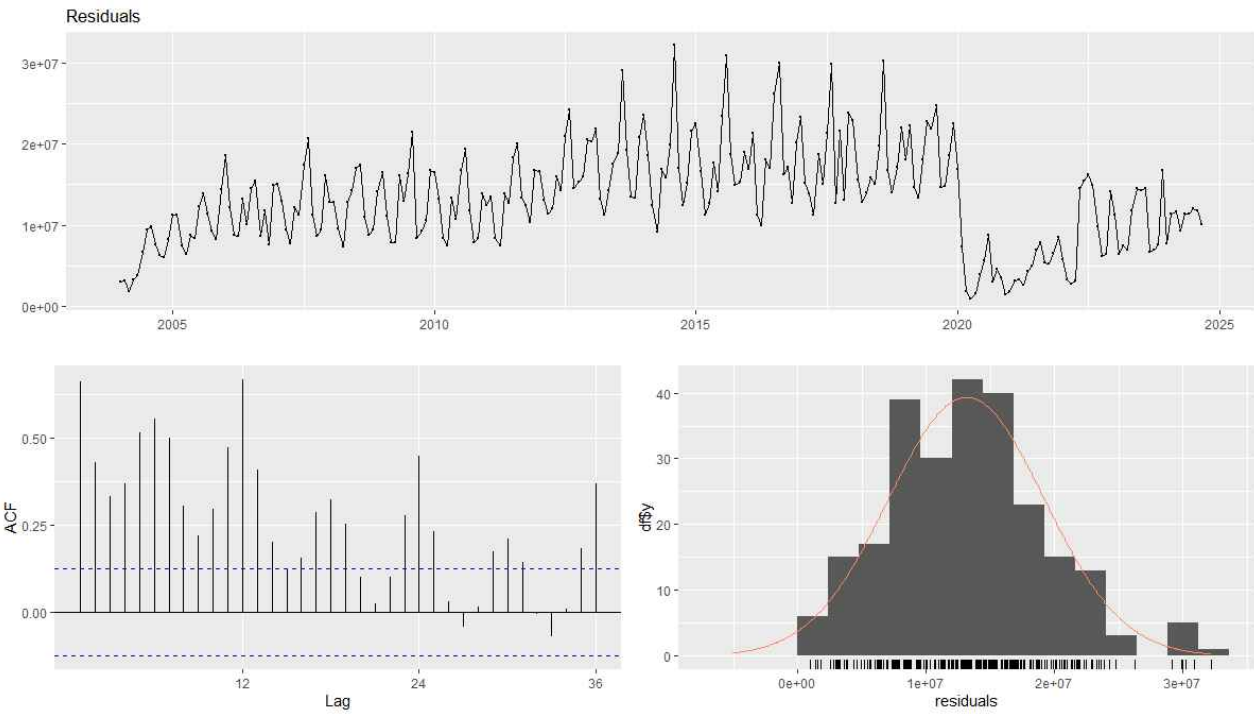
As shown in the above <Figure 1>, a total of 249 units from January 2004 to September 2024 are the data on the number of people, and there are no missing values.



<Figure 2>

```
> checkresiduals(rowdata)

        Ljung-Box test

data:  Residuals
Q* = 884.87, df = 24, p-value < 2.2e-16

Model df: 0.   Total lags used: 24
```

<Figure 3>



<Figure 4>

The graph at the top of <Figure 4> is the time series plot of raw data. In addition, various plots such as ACF and PACF (<Figure 2>, <Figure 3>) can also be looked at. The time series plot reveals a seasonal pattern characterized by regular increases and decreases. And above all, we can see that the numbers drop sharply around January 2020. This overlaps with the time when the COVID-19 virus infected people appeared in Korea, and it is noticeable that the figure is low until about the first quarter of 2022. Except for the sharp drop in the numbers in early 2020, the overall trend can be said to be increasing.

ACF can say that correlation between residuals exists to the extent that sparks are observed in almost all lags. In addition, it can be assumed that seasonality exists from the observation of a specific pattern in which values increase and decrease amid a gradually decreasing trend. PACF is better than ACF, but it can be said that correlation exists when considering that the spark still exists.

The Ljung-Box test results indicate a significantly small p-value. “The Ljung-Box test” tests the existence of correlation between residuals. When the null hypothesis is rejected like <Figure 3>, it can be said that correlation exists between residuals.

(2) Possibly transferring the data

To stabilize variance, to make closer to the normal distribution, data is often transferred. The Box-Cox transformation is considered one of the most representative methods.

When looking at the plot at the lower right side of <Figure 4>, it can be seen that the histograms of the residuals show a pattern similar to the normal distribution. And above all, when looking at the time series plot, it seems difficult to say that there is a change in the variance as t increases. In other words, it can be said that data transferring is not required to make stable variance and normal distribution.

(3) Unit-root test

Next, it is necessary to find out whether the unit-root exists in the data. The existence of a unit-root means that differentiation is necessary, which can be confirmed through the DF (Dickey-Fuller) test, KPSS test, and PP (Philips-Perron) test. In this project, we intend to test using the DF test. In the case of the DF test, if the null hypothesis is rejected, it can be said that the random walk exists in the model, which leads to the interpretation of the existence of unit-root and the need for differentiation. In fact, the DF test is conducted on the above data,

```
> adf.test(rowdata)

        Augmented Dickey-Fuller Test

data:  rowdata
Dickey-Fuller = -2.4411, Lag order = 6, p-value = 0.3901
alternative hypothesis: stationary
```

<Figure 5>

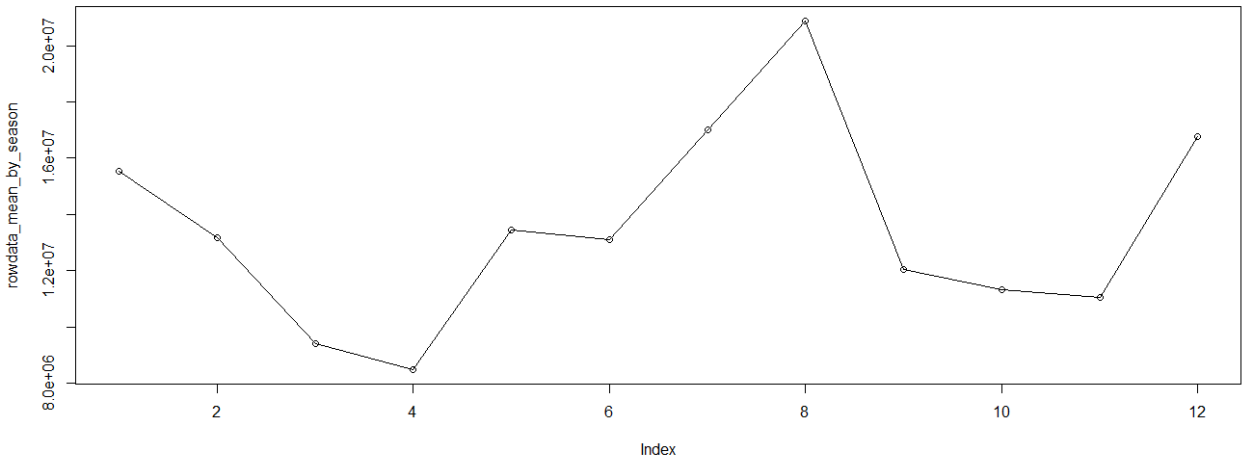
such a result is derived. The null hypothesis cannot be rejected because the p-value is a large value at the significance level of 0.05. In other words, the null hypothesis is accepted, which can be said to have a unit-root. Based on this conclusion, differentiation is performed, and the data is also tested once more because unit-root may exist.

```
> adf.test(diff(rowdata))

        Augmented Dickey-Fuller Test

data:  diff(rowdata)
Dickey-Fuller = -7.676, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary
```

<Figure 6>

This is the result of the DF test with the data from which the differentiation was performed once. This time, the p-value is very small at 0.01, which makes it possible to reject the null hypothesis at the significance level of 0.05. In other words, it can be said that there is no unit-root in the above data, which concludes that no additional differentiation is required.

(4) Seasonality



<Figure 7>

<Figure 7> is a graph of the seasonal average. In other words, seasonal trends can be identified, which leads to confirmation of seasonality. As can be seen from the graph above, it recorded the lowest value in April and the highest value in August. In addition, it can be seen that the values are quite high in December and January, which are the end of the year and the beginning of the year. In this plot, it can be said that seasonality exists because a specific pattern is observed that decreases in spring and autumn and then increases in summer and winter.

(5) Apply ARIMA Model

So far, we have looked at the characteristics of row data. Through (1), it was found that correlation between the residuals exists, and through (2), it was shown that separate data transferring is not required. Through (3) and (4), it was confirmed through DF test and season plot that unit-root and seasonality exist. Based on these results, a specific ARIMA model can be obtained.

```
> fit1=auto.arima(rowdata);fit1
Series: rowdata
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
          ma1     sma1
      -0.5520  -0.6654
s.e.   0.0608   0.0630

sigma^2 = 8.835e+12:  log likelihood = -3855.11
AIC=7716.22   AICc=7716.32   BIC=7726.61
```

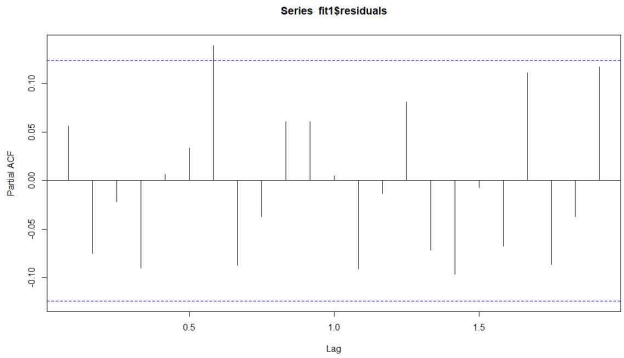
<Figure 8>

The model calculated using the auto.arima function is ARIMA(0,1,1)(0,1,1)[12]. When ARIMA(p,d,q)(P,D,Q)[s], it can be said that d is 1 reflects the result of (3). In the case of D, if seasonality exists, it is differentiation to remove it, so it can be judged that the interpretation

of (4) is correct.

(6) Diagnostics

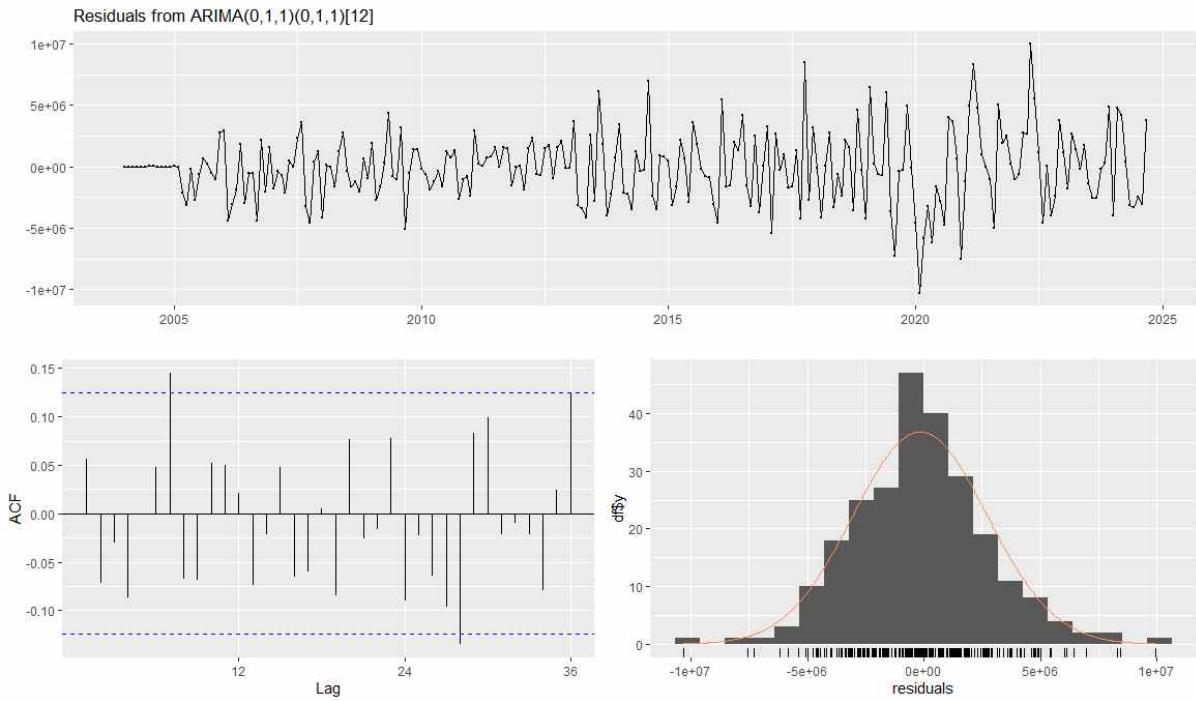
So, for the ARIMA model obtained in (5), can we say that it is appropriate data to describe the data?



<Figure 9>

```
> checkresiduals(fit1)

        Ljung-Box test

data:  Residuals from ARIMA(0,1,1)(0,1,1)[12]
Q* = 26.072, df = 22, p-value = 0.2486

Model df: 2.   Total lags used: 24
```

<Figure 10>



<Figure 11>

When looking at the reserve plot, no specific pattern is observed. <Figure 10> is the result of analyzing the ARIMA model using the checkresiduals function. Unlike in (1), the result of the Ljung-Box test shows that the p-value is sufficiently large compared to the significance level of 0.05, which accepts the null hypothesis. In other words, it can be said that there is no correlation between the reserves. In addition, even if you look at the PACF plot in <Figure 9> and the ACF plot in <Figure 11>, it can be seen that the spots are significantly reduced, unlike the ACF plots in (1). When looking at the histograms of the reserves, the distribution similar to the normal distribution also supports that ARIMA(0,1,1)(0,1,1)[12] is a suitable model for explaining data.

(7) ARIMA(0,1,1)(0,1,1)[12]

Through the analysis in (6), it was found that the model derived from the auto.arima function was statistically significant in explaining the monthly movie theater audience data. If the model is expressed in an equation with coefficient based on <Figure 8>,

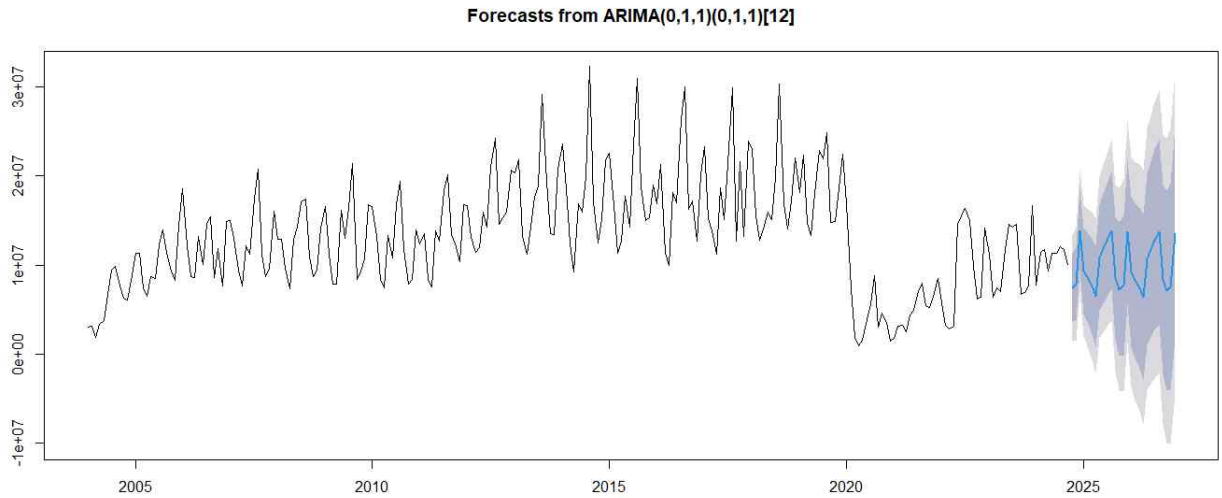
$$y_t = y_{t-1} + y_{t-12} - y_{t-13} + \epsilon_t - 0.552\epsilon_{t-1} - 0.6654\epsilon_{t-12} + 0.3673\epsilon_{t-13}$$

$$\epsilon_t \sim IID(0, 8.835e+12)$$

It can be written with this(for the coefficient of 0.3673, multiplied by the coefficient of ma1, sma1).

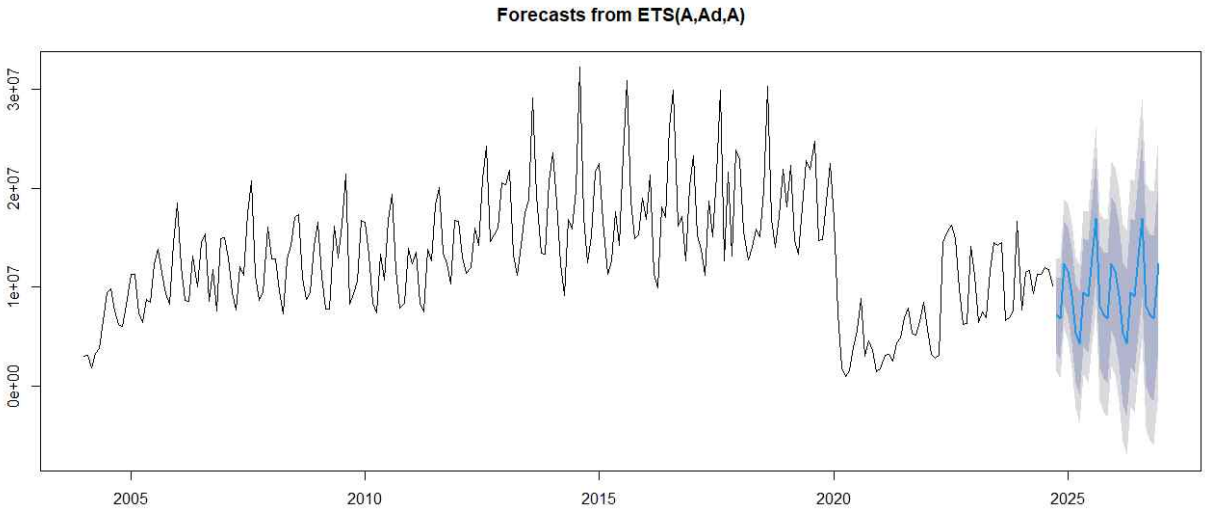
(8) forecasting

Based on the finally calculated model, viewer data can be interpreted. Based on this, data up to December 2026 is forcasting to understand trends in the film industry represented by the number of visitors in the future.



<Figure 12>

The plot above is the result of forcasting based on the finally derived model. As a result of forcasting, it can be seen that it is not possible to recover to the same values as before 2020. It is difficult to see not only the recovery but also the increasing trend. If so, to what extent can the results of this forcasting be trusted? For comparison, we intend to use the results of forcasting through a model with exponential smoothing.



<Figure 13>

This is the result of forcasting through ETS(A, Ad, A) proposed by R. Here, among the three factors, the first A means Additive of Residuals, the second Ad means Additive Damped of Trend, and the last A means Additive of Seasonal Component. Rough predictions are similar in both models, but there are differences in detailed values, in that there is no noticeable trend. To compare two models, comparing RMSE may be a method.

```
> accuracy(forecast_ARIMA)
                    ME    RMSE     MAE      MPE     MAPE      MASE       ACF1
Training set -159087.7 2881495 2161161 -6.58647 26.31795 0.6823097 0.05625686
> accuracy(forecast_ETS)
                    ME    RMSE     MAE       MPE     MAPE     MASE       ACF1
Training set -57795.68 2796752 2166807 -7.529326 28.03252 0.684092 0.09291494
```

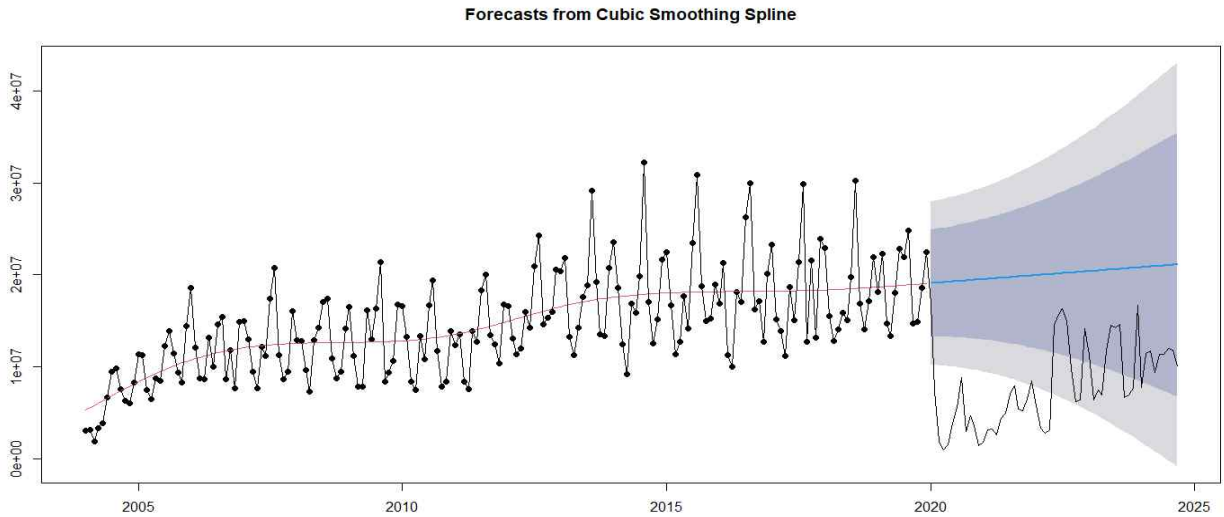
<Figure 14>

Forcast_ARIMA corresponds to the model of <Figure 12> and forcast_ETS corresponds to <Figure 13>. When comparing the RMSE of the two models, it can be seen that the case of ETS is smaller. Therefore, among the two models, it can be said that forcasting through ETS (A, Ad, A) has better performance, but it is judged that it is not too difficult to predict the forcast value of <Figure N> because the analysis was conducted using the ARIMA model in this project and there is no significant difference in the forcasting values between the two models.

(9) The Impact of COVID-19 on the Movie Theater Industry

As of January 2020, when the number of audiences decreased sharply due to COVID-19, the data will be split and analyzed. In other words, by training the data from December 2019, we want to predict the data from 2020 to September 2024. Compare the predicted value with the actual data to see what difference is made. However, a lot of attention is required in this analysis. It should be assumed that COVID-19 is the only factor that differs from the previous

values, even though the data for January 2020 has rapidly decreased (t despite increasing). It should also be stated that the trend is non-linear because it will be forecast using a spline. The results of the analysis after going through these assumptions are as follows.



<Figure 15>

As a result of forcasting from January 2020 using the spline function, it shows a gradually increasing trend. In other words, it is different from the actual data assuming that it decreased sharply due to the influence of COVID-19.

4. Results

This is a conclusion about the trends and future of the movie theater industry examined by the number of audiences. It was found by looking at tests and plots that the data have correlation, unit-root, and seasonality between the residuals. Based on this, the basis for the model of ARIMA(0,1,1)(0,1,1)[12], a model derived using the auto.arima function, was secured, and the evidence that this model is also suitable for explaining the data could be concluded through plots and tests. It can also be seen that when forcasting with this model, the increasing trend does not appear clearly.

In conclusion, the number of moviegoers demonstrates a general upward trend, with clear seasonality marked by increases in summer and winter and declines in spring and fall. Above all, as of January 2020, you can see a plot with a different pattern based on the time when COVID-19 appeared in Korean society. This can be visualized through a gap with actual data when forcasting was conducted as of January 2020, as in <Figure N>. In other words, it can be said that the pandemic has affected the movie theater industry, assuming that COVID-19 is the only factor that caused the sharp decline in data in January 2020.

5. Discussion

The number of moviegoers can be divided into January 2020 before and after the COVID-19 virus appeared in Korean society. Until then, the size of the entire movie theater industry had been gradually increasing amid a pattern in which the number of visitors increased and decreased depending on the season. However, identifying a clear recovery pattern since the severe downturn in January 2020 remains challenging. The disappearance of distinct seasonal patterns has also changed since then, and it is difficult to find a noticeable increase in the forecast for the next two years.

However, there are some limitations of this project and some additional things to discuss.

(1) First of all, the number of moviegoers is not directly related to the movie theater industry. The number of moviegoers in Korea is always mentioned, and it is true that it accounts for the largest portion of movie theater sales. However, this data is not directly linked to the industry, and there are other factors, such as an increase in ticket prices per seat in a movie theater and various sales strategies.

(2) The coronavirus may not have been the only factor that contributed to the reason why the number of spectators who gave a sharp decline due to COVID-19 does not easily remain the same even after the end of the pandemic. The growth of the OTT industry is a prime example, as previously mentioned in introduction. It is desirable to compare the recent trends in the OTT industry over the past five years with those in the movie theater industry represented by the number of audiences to further discuss what kind of relationship exists between the two industries and whether they can be said to affect each other.